

- d) No student in your class has a cat, a dog and a ferret:
 $\neg \exists x (C(x) \wedge D(x) \wedge F(x))$
- e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet: $\exists x C(x) \wedge \exists x D(x) \wedge \exists x F(x)$
- 13) Determine the truth value of each of these statements if the domain consists of all integers.

- a) $\forall n (n+1) > n$. Let $\neg \forall n (n+1) > n$ be True. By De Morgan's law of quantifiers, $\neg \forall n (n+1) > n \equiv \exists n \neg (n+1) > n$. $\therefore \exists n \neg (n+1) > n$ is True or, $\exists n, (n+1) \leq n \Rightarrow \exists n, 1 \leq 0$. But this is False.
 $\therefore \neg \forall n (n+1) > n$ is False and hence by the defn. of negation and double-negation law, $\forall n (n+1) > n$ is True.
- b) $\exists n (2n = 3n)$. Let, $\exists n (2n = 3n)$ be True. Let, $2x = 3x$.
 $\Rightarrow 2(3-2) = 0 \Rightarrow x = 0$. $\therefore$ For $x = 0$, $\exists n (2n = 3n)$ is True
- c) $\exists n (n = -n)$ Let, $\exists n (n = -n)$ be True. $\therefore n = -n \Rightarrow (n+n) = 0 \Rightarrow 2n = 0$
 $\Rightarrow n = 0$

- $\therefore$ If $n = 0$, $\exists n (n = -n)$ is True.
- d) $\forall n (3n < 4n)$ Let, $\forall n (3n < 4n)$ be False. By defn. of negation, $\neg \forall n (3n < 4n)$ is True. By De Morgan's law of quantifiers, $\exists n \neg (3n < 4n)$ is True, or $\exists n (3n \geq 4n)$ is True.
 Let, $3x \geq 4x$ for some x . $\therefore 0 \geq x$ for some x .
 This holds for all negative integers.

$\therefore \forall n (3n < 4n)$ is False

- 15) Determine the truth value of each of these statements if the domain for all variables consists of all integers.
- a) $\forall n (n^2 > 0)$. We know, the square of any integer, positive or negative is always positive. This holds true for any integer.
 $\therefore \forall n (n^2 > 0)$ is True.